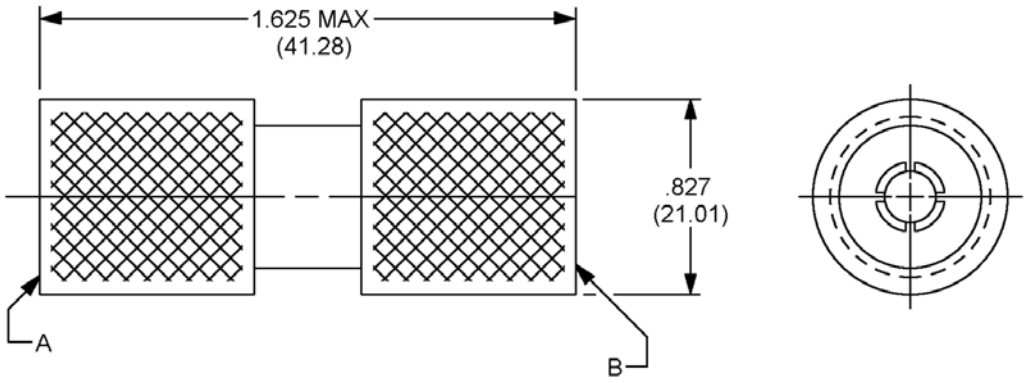
INCH-POUND
MIL-PRF-55339/5A
w/AMENDMENT 1
16 February 2018
SUPERSEDING
MIL-PRF-55339/5A
10 January 2005

PERFORMANCE SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, IN-LINE,
(WITHIN SERIES N PLUG TO SERIES N PLUG), CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.




| Reference | Series | Contact |
|---|---|---|
| A and B | N | Pin |

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. All undimensioned pictorial representations are for reference purposes only.
3. Interface dimensions shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A

FSC 5935

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DESIGN AND CONSTRUCTION:

General configuration: See figure 1.

Impedance: 50 ohms, nominal.

Working voltage:

Sea level: 1,000 Vrms.

70,000 feet (4.437 kPa): 250 Vrms.

Frequency range: 0 to 11 GHz.

Temperature range: -65° to +165°C.

PERFORMANCE (installation torque of 6 to 10 in. lb (0.68 to 1.13 Nm)):

Dimensions: See figure 1 and MIL-STD-348.

Center contact retention:

Axial force: 6 lb minimum (26.69 N).

Torque: Not applicable.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 in. lb (0.68 Nm), maximum.

Coupling proof torque: 15 in. lb (1.69 Nm), minimum.

Mating characteristics:

Center contact (socket): Not applicable.

Outer contact:

Minimum test ring ID: .316 inch (8.03 mm), maximum.

Ring finish: 16 microinches (0.406 µm).

Insertion force: 25 lb (111.20 N), maximum.

Insertion depth: .093 inch (2.36 mm), minimum.

Number of insertions: 1.



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Maximum test ring ID: .324 inch (8.23 mm), minimum.

Test ring finish: 16 microinches (0.406 µm).

Insertion depth: .031 inch (0.79 mm), maximum.

Number of insertions: 1.

Permeability: <2.0

Seal:

Hermetic: Not applicable.

Pressurized: Not applicable.

Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR: 1.30, maximum .5 to 11 GHz.

RF leakage (total): -90 dB, minimum, 2 to 3 GHz.

RF insertion loss: .15 dB, maximum, 9 GHz (.050 √F (GHz) dB maximum tested at 9 GHz).

Durability: 500 cycles minimum at 12 cycles/minimum, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding:

Test voltage: 2,500 Vrms, minimum (sea level).

Contact resistance (milliohms, maximum).

| Contact | Initial | After environment |
|---|---|---|
| Center | 1.5 | 2.0 |
| Outer | 0.3 | N/A |
| Outer (-70001) | 0.6 | N/A |

Vibration, high frequency:

Interruptions: 1 µs, maximum.

Shock: Test condition I.

Thermal shock: Test condition C.

Moisture resistance: 200 megohms, minimum



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Corona level:

Voltage: 500 V, minimum.

Altitude: 70,000 feet (4.437 kPa), minimum.

RF high potential withstanding voltage:

RF voltage: 1,500 Vrms, minimum.

Frequency: 5 MHz, minimum.

Salt spray (corrosion): Applicable.

Coupling mechanism retention force: 100 lb (444.82 Nm), minimum.

Part or Identifying Number (PIN):

M55339/05-00057 or:

M55339/05-70001 **CAUTION: THIS PART HAS A NICKEL PLATED BODY AND IS NOT FOR USE IN APPLICATIONS WHERE PASSIVE INTERMODULATION GENERATION (PIM) MAY BE A CONCERN.**

TABLE I. Cross reference of PINs.

| PIN | Superseded PIN or type designation 1/ |
|---|---|
| M55339/05-00057 | MS15507<br>UG-57 B/U |

1/ The superseded PIN or the type designation is for cross-reference only. Where a superseded PIN or type designation is not given, none was assigned or will be assigned. The PIN: M55339/05-00057 shall be used in all cases for marking and identifying the adapter.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to MIL-PRF-55339, this document references the following:

MIL-STD-348



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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2018-029)

Review activities:
Army - AR, AT, EA, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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